

```

In [12]: import numpy as np
import warnings
warnings.filterwarnings('ignore')

#Seasons
Seasons = ["2016", "2017", "2018", "2019", "2020", "2021", "2022", "2023", "2024", "2025"]
Sdict = {"2016":0, "2017":1, "2018":2, "2019":3, "2020":4, "2021":5, "2022":6, "2023":7

#Players
Players = ["Sachin", "Rahul", "Smith", "Sami", "Pollard", "Morris", "Samson", "Dhoni", "
Pdict = {"Sachin":0, "Rahul":1, "Smith":2, "Sami":3, "Pollard":4, "Morris":5, "Samson"

#Salaries
Sachin_Salary = [15946875, 17718750, 19490625, 21262500, 23034375, 24806250, 25244493,
Rahul_Salary = [12000000, 12744189, 13488377, 14232567, 14976754, 16324500, 18038573, 1
Smith_Salary = [4621800, 5828090, 13041250, 14410581, 15779912, 14500000, 16022500, 175
Sami_Salary = [3713640, 4694041, 13041250, 14410581, 15779912, 17149243, 18518574, 1945
Pollard_Salary = [4493160, 4806720, 6061274, 13758000, 15202590, 16647180, 18091770, 19
Morris_Salary = [3348000, 4235220, 12455000, 14410581, 15779912, 14500000, 16022500, 17
Samson_Salary = [3144240, 3380160, 3615960, 4574189, 13520500, 14940153, 16359805, 1777
Dhoni_Salary = [0, 0, 4171200, 4484040, 4796880, 6053663, 15506632, 16669630, 17832627, 1
Kohli_Salary = [0, 0, 0, 4822800, 5184480, 5546160, 6993708, 16402500, 17632688, 18862875
Sky_Salary = [3031920, 3841443, 13041250, 14410581, 15779912, 14200000, 15691000, 17182

#Matrix
Salary = np.array([Sachin_Salary, Rahul_Salary, Smith_Salary, Sami_Salary, Polla

#Games
Sachin_G = [80, 77, 82, 82, 73, 82, 58, 78, 6, 35]
Rahul_G = [82, 57, 82, 79, 76, 72, 60, 72, 79, 80]
Smith_G = [79, 78, 75, 81, 76, 79, 62, 76, 77, 69]
Sami_G = [80, 65, 77, 66, 69, 77, 55, 67, 77, 40]
Pollard_G = [82, 82, 82, 79, 82, 78, 54, 76, 71, 41]
Morris_G = [70, 69, 67, 77, 70, 77, 57, 74, 79, 44]
Samson_G = [78, 64, 80, 78, 45, 80, 60, 70, 62, 82]
Dhoni_G = [35, 35, 80, 74, 82, 78, 66, 81, 81, 27]
Kohli_G = [40, 40, 40, 81, 78, 81, 39, 0, 10, 51]
Sky_G = [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]

#Matrix
Games = np.array([Sachin_G, Rahul_G, Smith_G, Sami_G, Pollard_G, Morris_G, Samson_G, Dhoni_G, Kohli_G, Sky_G])

#Points
Sachin_PTS = [2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133, 83, 782]
Rahul_PTS = [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154]
Smith_PTS = [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743]
Sami_PTS = [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112, 966]
Pollard_PTS = [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297, 646]
Morris_PTS = [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281, 928]
Samson_PTS = [1258, 1104, 1684, 1781, 841, 1268, 1189, 1186, 1185, 1564]
Dhoni_PTS = [903, 903, 1624, 1871, 2472, 2161, 1850, 2280, 2593, 686]
Kohli_PTS = [597, 597, 597, 1361, 1619, 2026, 852, 0, 159, 904]
Sky_PTS = [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]

#Matrix
Points = np.array([Sachin_PTS, Rahul_PTS, Smith_PTS, Sami_PTS, Pollard_PTS, Morris_PTS, Samson_PTS, Dhoni_PTS, Kohli_PTS, Sky_PTS])

```

```

In [13]: Salary

```

```
Out[13]: array([[15946875, 17718750, 19490625, 21262500, 23034375, 24806250,
                25244493, 27849149, 30453805, 23500000],
               [12000000, 12744189, 13488377, 14232567, 14976754, 16324500,
                18038573, 19752645, 21466718, 23180790],
               [ 4621800,  5828090, 13041250, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3713640,  4694041, 13041250, 14410581, 15779912, 17149243,
                18518574, 19450000, 22407474, 22458000],
               [ 4493160,  4806720,  6061274, 13758000, 15202590, 16647180,
                18091770, 19536360, 20513178, 21436271],
               [ 3348000,  4235220, 12455000, 14410581, 15779912, 14500000,
                16022500, 17545000, 19067500, 20644400],
               [ 3144240,  3380160,  3615960,  4574189, 13520500, 14940153,
                16359805, 17779458, 18668431, 20068563],
               [      0,      0,  4171200,  4484040,  4796880,  6053663,
                15506632, 16669630, 17832627, 18995624],
               [      0,      0,      0,  4822800,  5184480,  5546160,
                6993708, 16402500, 17632688, 18862875],
               [ 3031920,  3841443, 13041250, 14410581, 15779912, 14200000,
                15691000, 17182000, 18673000, 15000000]])
```

In [14]: Games

```
Out[14]: array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
               [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
               [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
               [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
               [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
               [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
               [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
               [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
               [40, 40, 40, 81, 78, 81, 39,  0, 10, 51],
               [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

In [15]: Points

```
Out[15]: array([[2832, 2430, 2323, 2201, 1970, 2078, 1616, 2133,  83, 782],
               [1653, 1426, 1779, 1688, 1619, 1312, 1129, 1170, 1245, 1154],
               [2478, 2132, 2250, 2304, 2258, 2111, 1683, 2036, 2089, 1743],
               [2122, 1881, 1978, 1504, 1943, 1970, 1245, 1920, 2112,  966],
               [1292, 1443, 1695, 1624, 1503, 1784, 1113, 1296, 1297,  646],
               [1572, 1561, 1496, 1746, 1678, 1438, 1025, 1232, 1281,  928],
               [1258, 1104, 1684, 1781,  841, 1268, 1189, 1186, 1185, 1564],
               [ 903,  903, 1624, 1871, 2472, 2161, 1850, 2280, 2593,  686],
               [ 597,  597,  597, 1361, 1619, 2026,  852,  0, 159,  904],
               [2040, 1397, 1254, 2386, 2045, 1941, 1082, 1463, 1028, 1331]])
```

In [16]: len(Games)

Out[16]: 10

In [17]: Games.shape

Out[17]: (10, 10)

In [18]: np.round(Salary / Games)

```
Out[18]: array([[ 199336.,  230114.,  237691.,  259299.,  315539.,  302515.,
                  435250.,  357040.,  5075634.,  671429.],
                [ 146341.,  223582.,  164492.,  180159.,  197063.,  226729.,
                  300643.,  274342.,  271731.,  289760.],
                [  58504.,   74719.,  173883.,  177908.,  207630.,  183544.,
                  258427.,  230855.,  247630.,  299194.],
                [  46420.,   72216.,  169367.,  218342.,  228694.,  222717.,
                  336701.,  290299.,  291006.,  561450.],
                [  54795.,   58619.,   73918.,  174152.,  185397.,  213425.,
                  335033.,  257057.,  288918.,  522836.],
                [  47829.,   61380.,  185896.,  187150.,  225427.,  188312.,
                  281096.,  237095.,  241361.,  469191.],
                [  40311.,   52815.,   45200.,   58643.,  300456.,  186752.,
                  272663.,  253992.,  301104.,  244739.],
                [    0.,    0.,   52140.,   60595.,   58499.,   77611.,
                  234949.,  205798.,  220156.,  703542.],
                [    0.,    0.,    0.,   59541.,   66468.,   68471.,
                  179326.,   inf,  1763269.,  369860.],
                [  40426.,   75322.,  255711.,  182412.,  204934.,  186842.,
                  320224.,  249014.,  345796.,  241935.]])
```

```
In [34]: Games[0]
```

```
Out[34]: array([80, 77, 82, 82, 73, 82, 58, 78,  6, 35])
```

```
In [35]: Games[0:5]
```

```
Out[35]: array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
                [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
                [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
                [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
                [82, 82, 82, 79, 82, 78, 54, 76, 71, 41]])
```

```
In [36]: Games[0,5]
```

```
Out[36]: np.int64(82)
```

```
In [37]: Games
```

```
Out[37]: array([[80, 77, 82, 82, 73, 82, 58, 78,  6, 35],
                [82, 57, 82, 79, 76, 72, 60, 72, 79, 80],
                [79, 78, 75, 81, 76, 79, 62, 76, 77, 69],
                [80, 65, 77, 66, 69, 77, 55, 67, 77, 40],
                [82, 82, 82, 79, 82, 78, 54, 76, 71, 41],
                [70, 69, 67, 77, 70, 77, 57, 74, 79, 44],
                [78, 64, 80, 78, 45, 80, 60, 70, 62, 82],
                [35, 35, 80, 74, 82, 78, 66, 81, 81, 27],
                [40, 40, 40, 81, 78, 81, 39,  0, 10, 51],
                [75, 51, 51, 79, 77, 76, 49, 69, 54, 62]])
```

```
In [38]: Games[1:2]
```

```
Out[38]: array([[82, 57, 82, 79, 76, 72, 60, 72, 79, 80]])
```

```
In [39]: Games[-3,-1]
```

```
Out[39]: np.int64(27)
```

```
In [40]: Pdict
```

```
Out[40]: {'Sachin': 0,  
          'Rahul': 1,  
          'Smith': 2,  
          'Sami': 3,  
          'Pollard': 4,  
          'Morris': 5,  
          'Samson': 6,  
          'Dhoni': 7,  
          'Kohli': 8,  
          'Sky': 9}
```

```
In [41]: Sdict
```

```
Out[41]: {'2016': 0,  
          '2017': 1,  
          '2018': 2,  
          '2019': 3,  
          '2020': 4,  
          '2021': 5,  
          '2022': 6,  
          '2023': 7,  
          '2024': 8,  
          '2025': 9}
```

```
In [42]: Pdict['Sachin']
```

```
Out[42]: 0
```

```
In [43]: Pdict['Rahul']
```

```
Out[43]: 1
```

```
In [46]: Sdict['2025']
```

```
Out[46]: 9
```

```
In [48]: Games[Pdict['Rahul']]
```

```
Out[48]: array([82, 57, 82, 79, 76, 72, 60, 72, 79, 80])
```

```
In [49]: Salary[Pdict['Rahul']][Sdict['2020']]
```

```
Out[49]: np.int64(14976754)
```

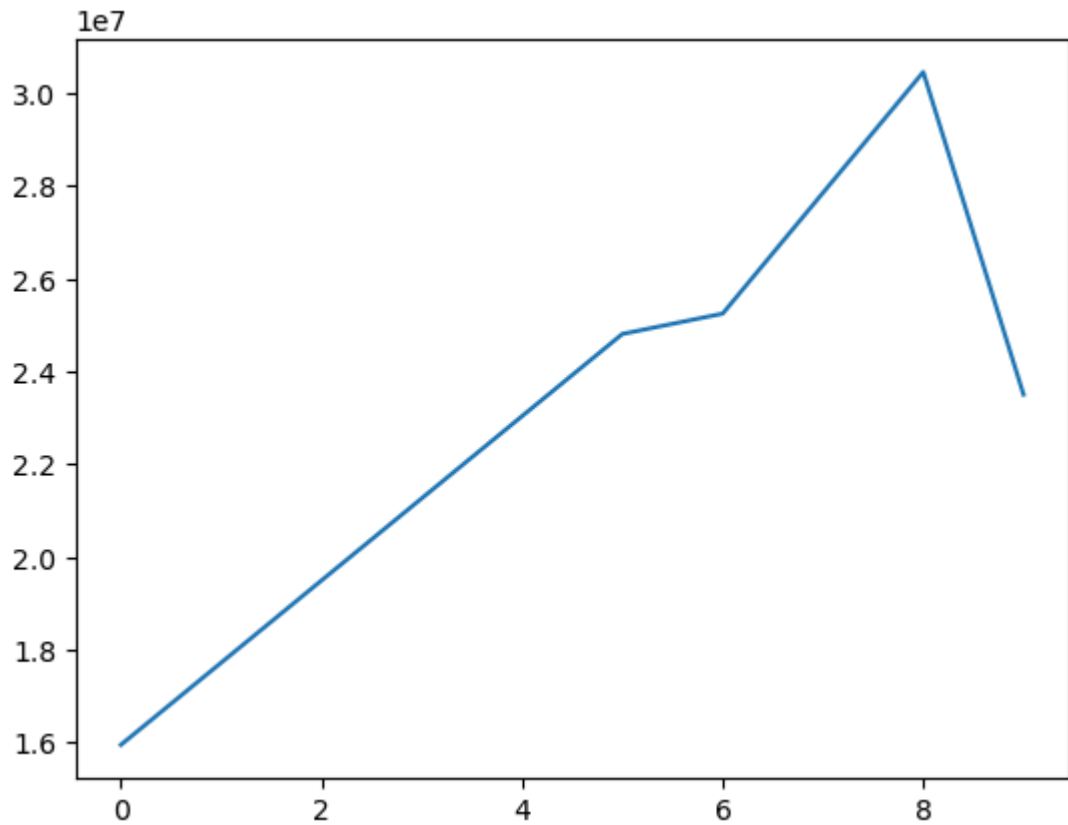
```
In [20]: import matplotlib.pyplot as plt  
%matplotlib inline
```

```
In [21]: Salary[0]
```

```
Out[21]: array([15946875, 17718750, 19490625, 21262500, 23034375, 24806250,  
               25244493, 27849149, 30453805, 23500000])
```

```
In [22]: plt.plot(Salary[0])
```

```
Out[22]: [<matplotlib.lines.Line2D at 0x2a5c2610050>]
```

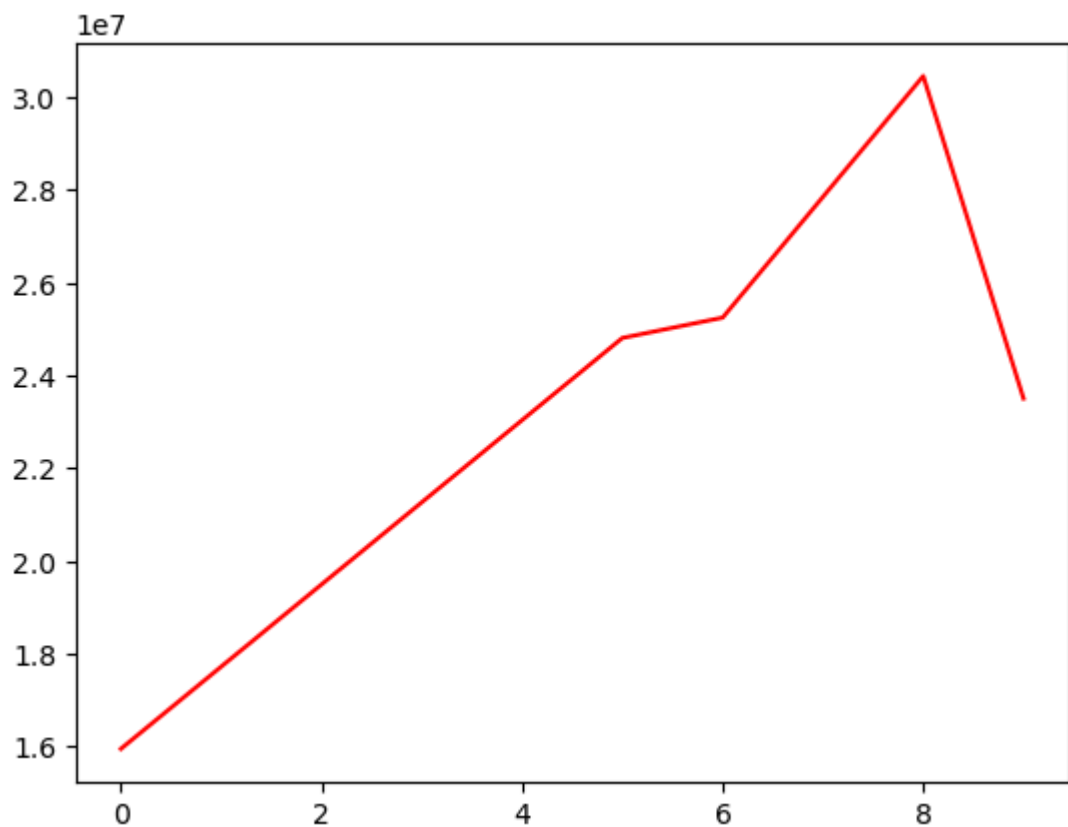


```
In [23]: Games[0]
```

```
Out[23]: array([80, 77, 82, 82, 73, 82, 58, 78, 6, 35])
```

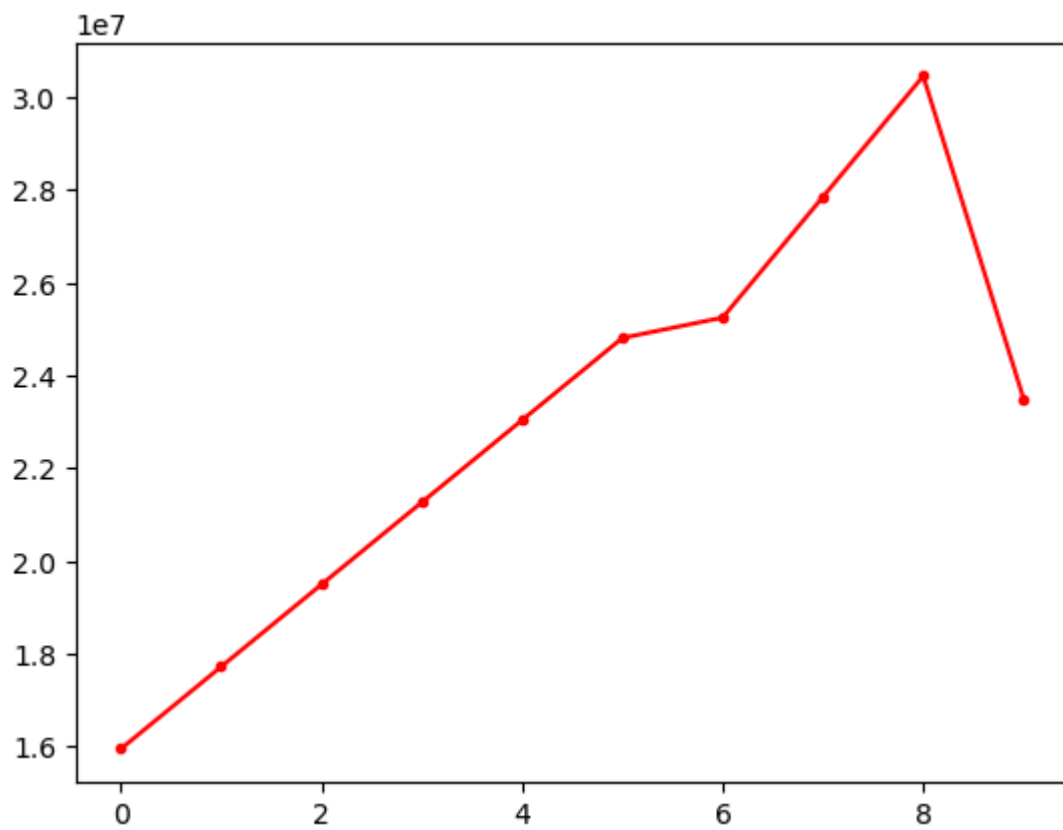
```
In [24]: plt.plot(Salary[0],color = 'red')
```

```
Out[24]: [<matplotlib.lines.Line2D at 0x2a5c36bca50>]
```



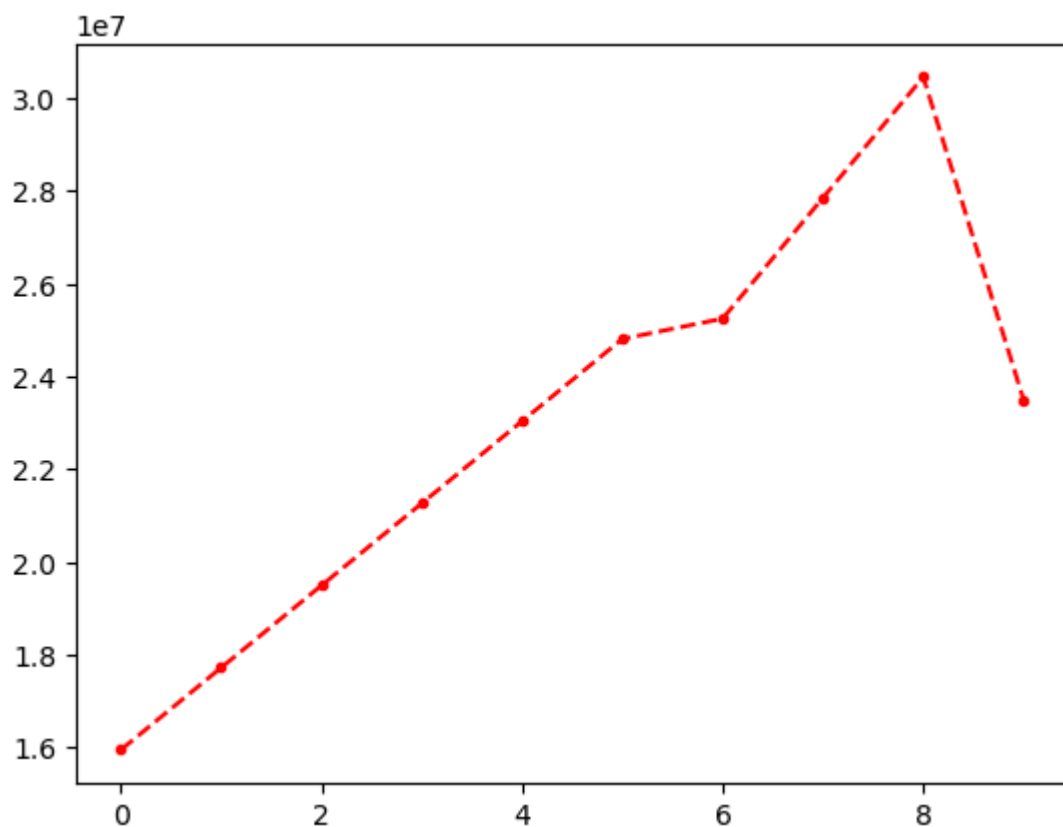
```
In [25]: plt.plot(Salary[0],color = 'r',marker= '.')
```

```
Out[25]: [<matplotlib.lines.Line2D at 0x2a5c388bb10>]
```



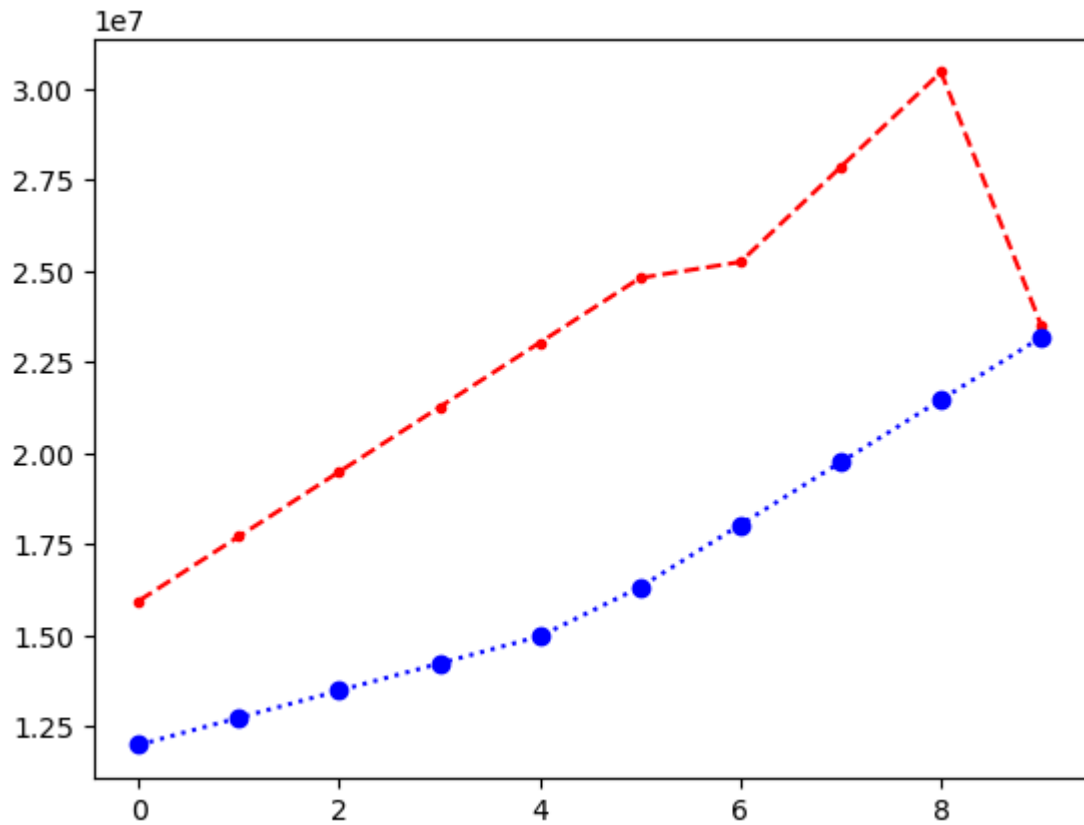
```
In [26]: plt.plot(Salary[0],color = 'r',marker= '.',ls = '--')
```

```
Out[26]: [<matplotlib.lines.Line2D at 0x2a5c390a0d0>]
```

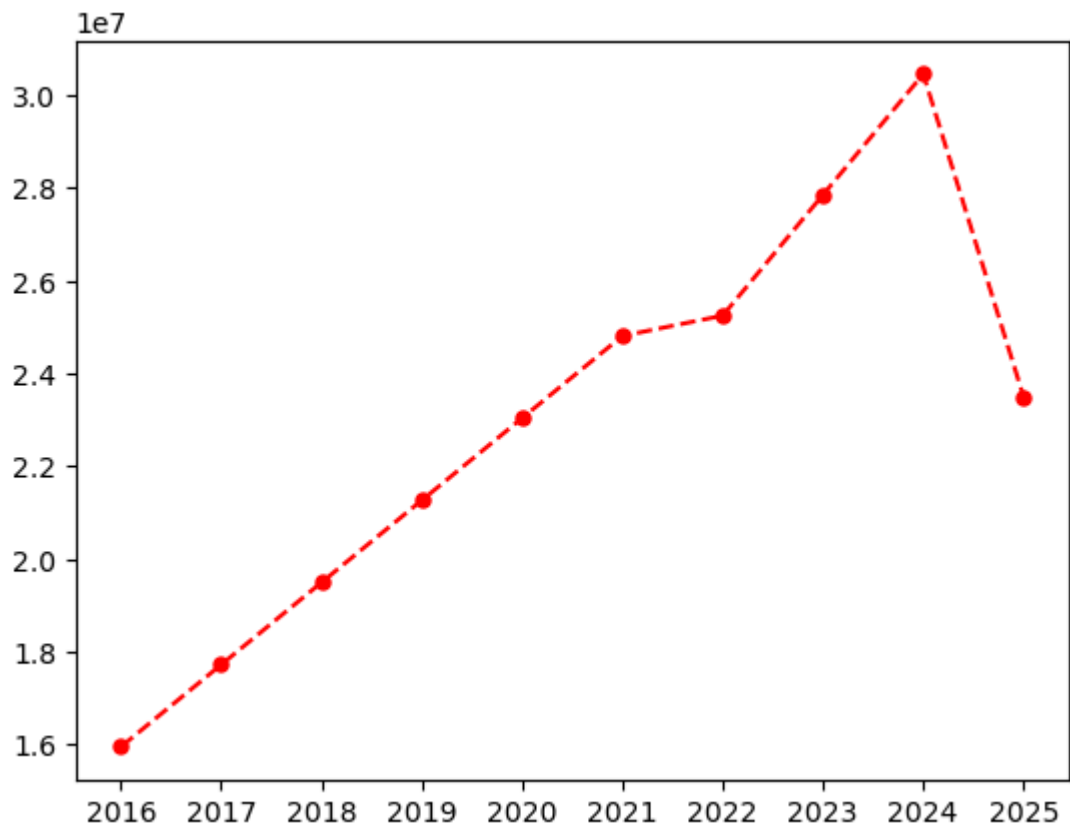


```
In [27]: plt.plot(Salary[0],color='r',marker='.',ls='--')  
plt.plot(Salary[1],color='b',marker='o',ls=':')
```

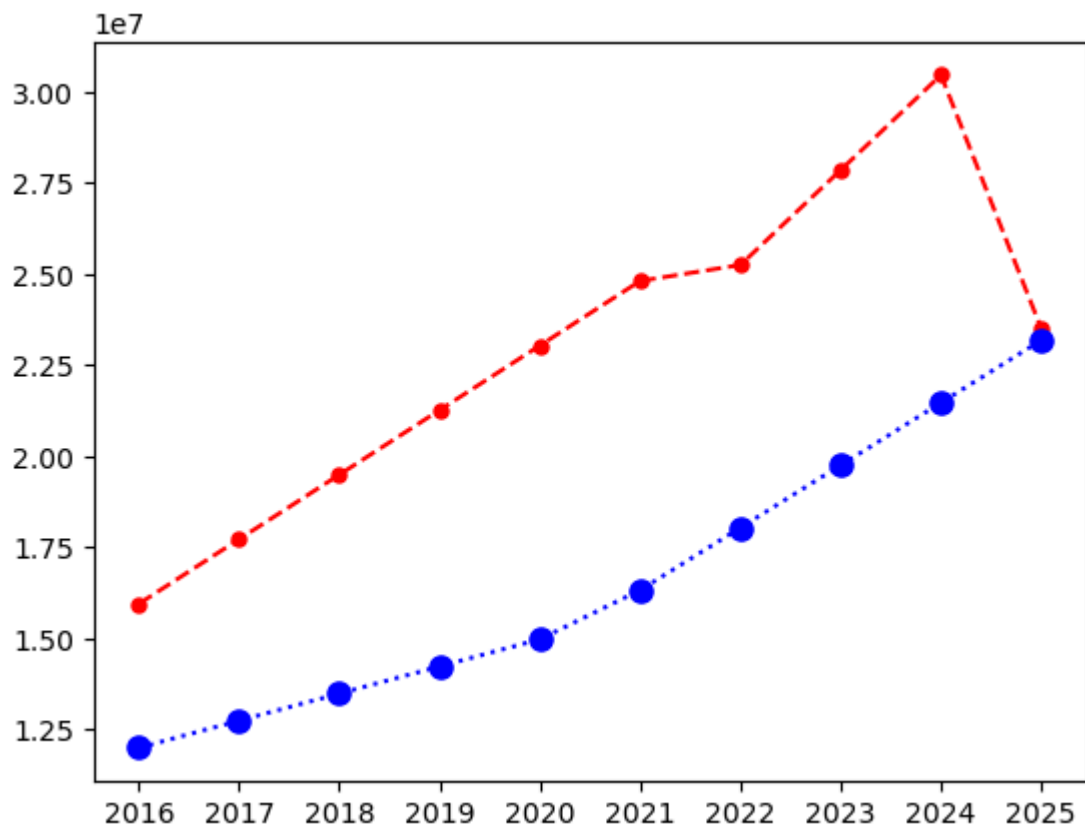
```
Out[27]: [<matplotlib.lines.Line2D at 0x2a5c3990910>]
```



```
In [28]: plt.plot(Salary[0],color='r',marker='.',ls='--',ms=10)  
plt.xticks(list(range(0,10)),Sdict)  
plt.show()
```



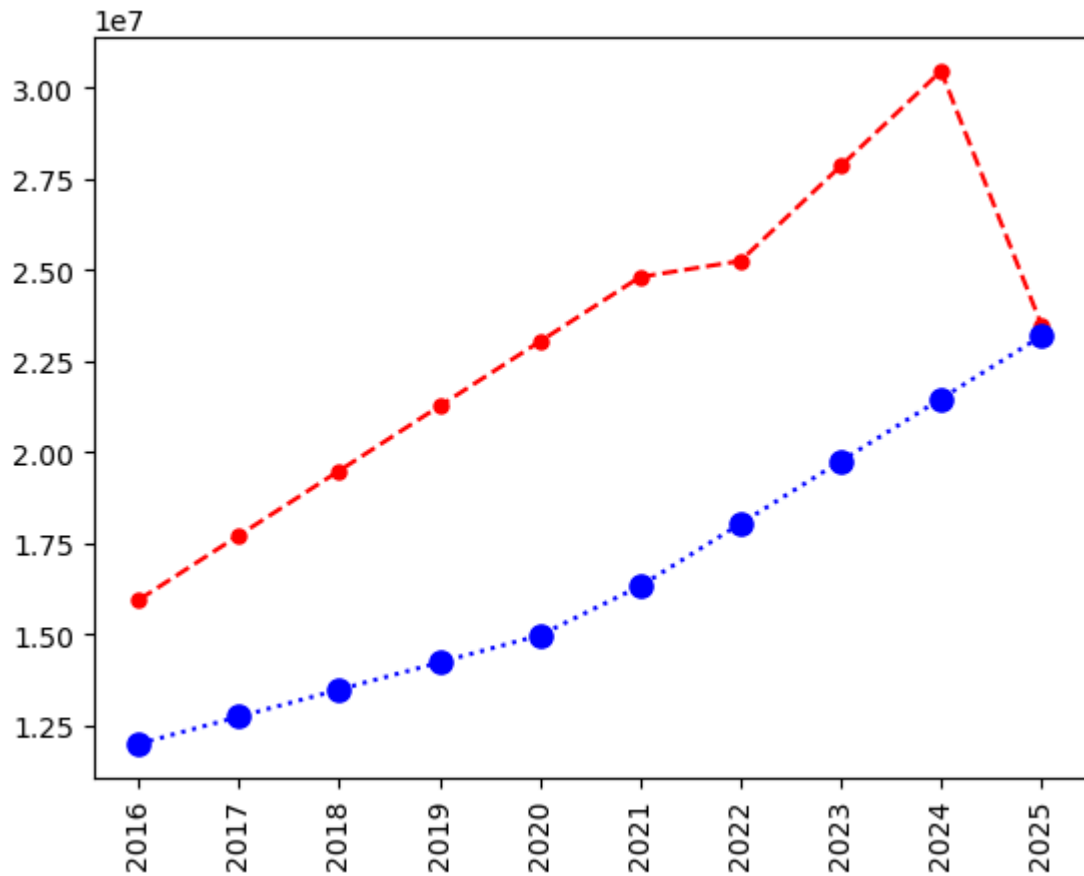
```
In [30]: plt.plot(Salary[0],color = 'r',marker= '.',ls = '--',ms=10)
plt.plot(Salary[1],color = 'b',marker= 'o',ls = ':',ms=8)
plt.xticks(list(range(0,10)),Sdict)
plt.show()
```



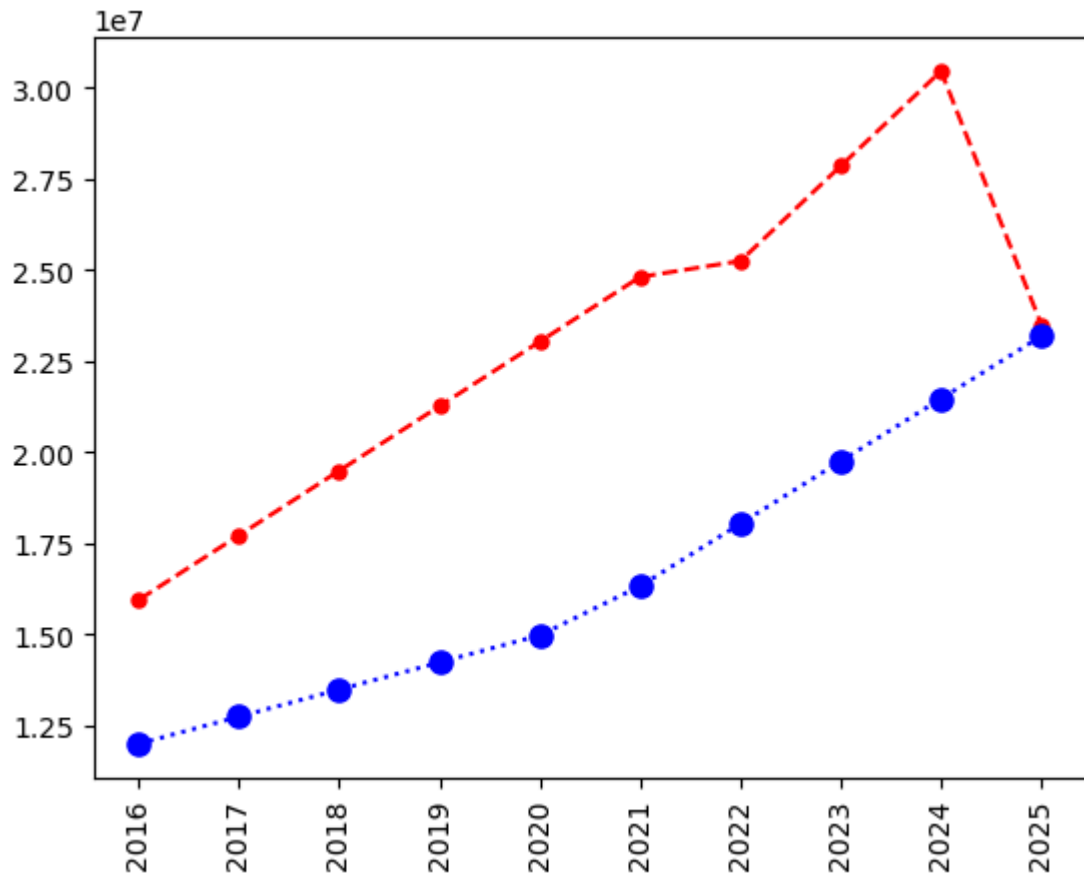
```
In [33]: plt.plot(Salary[0],color = 'r',marker= '.',ls = '--',ms=10)
plt.plot(Salary[1],color = 'b',marker= 'o',ls = ':',ms=8)
```



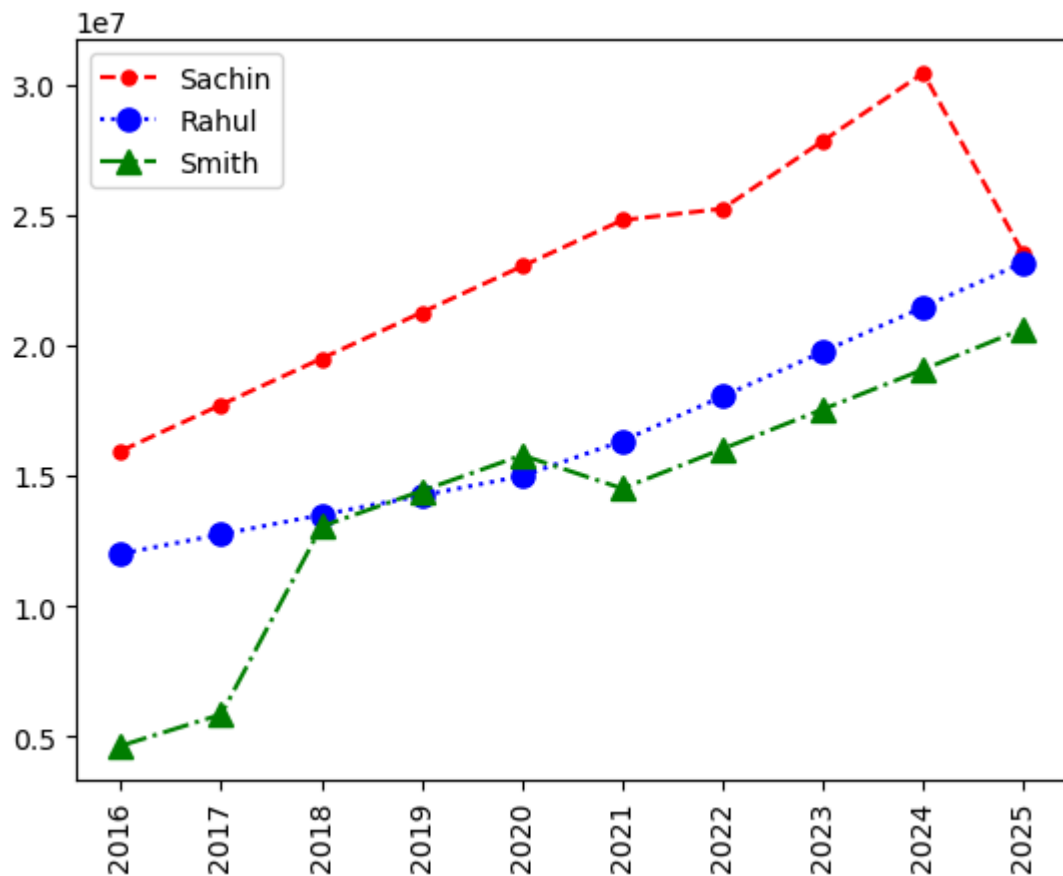
```
plt.xticks(list(range(0,10)),Sdict,rotation='vertical')  
plt.show()
```



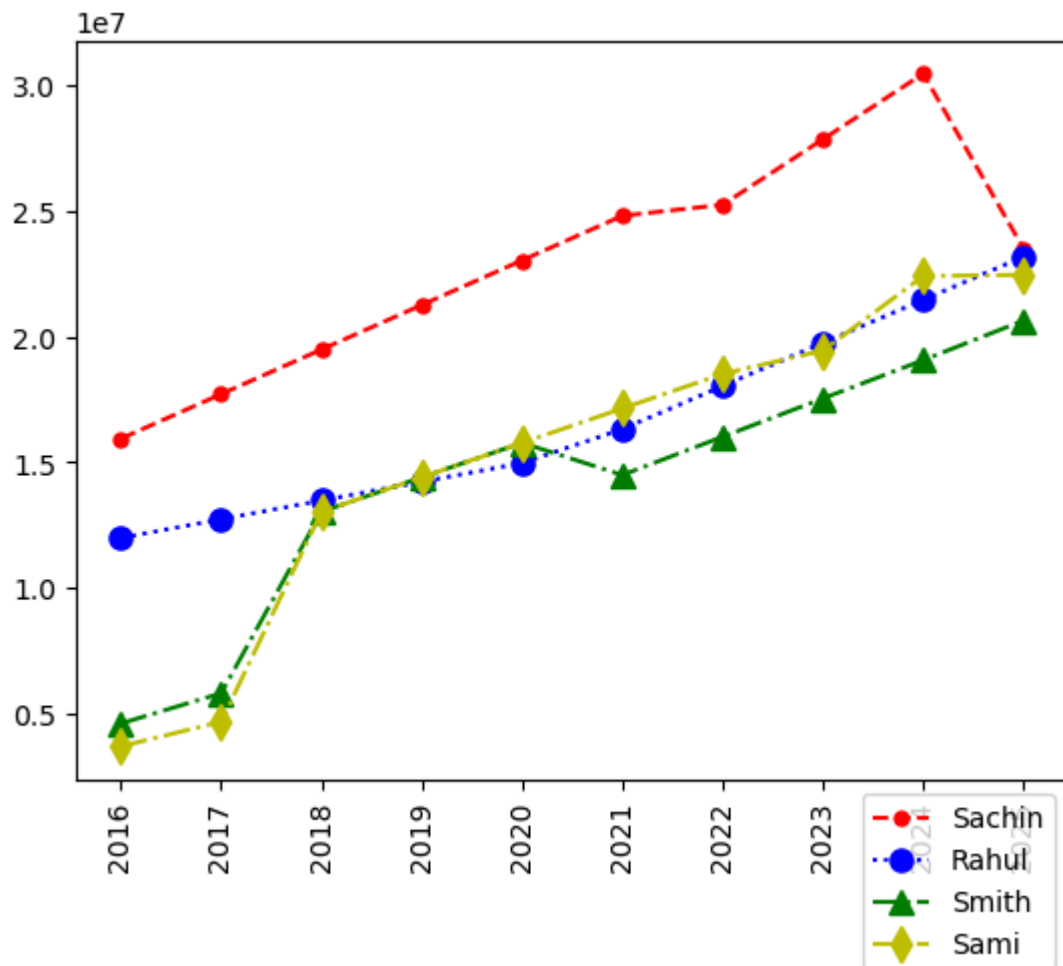
```
In [50]: plt.plot(Salary[0],color = 'r',marker= 'o',ls = '--',ms=10,label = Players[0])  
plt.plot(Salary[1],color = 'b',marker= 'o',ls = ':',ms=8,label = Players[1])  
plt.xticks(list(range(0,10)),Sdict,rotation='vertical')  
plt.show()
```



```
In [54]: plt.plot(Salary[0],color = 'r',marker= '.',ls = '--',ms=10,label = Players[0])
plt.plot(Salary[1],color = 'b',marker= 'o',ls = ':',ms=8,label = Players[1])
plt.plot(Salary[2],color = 'g',marker= '^',ls = '-.',ms=8,label = Players[2])
plt.xticks(list(range(0,10)),Sdict,rotation='vertical')
plt.legend()
plt.show()
```



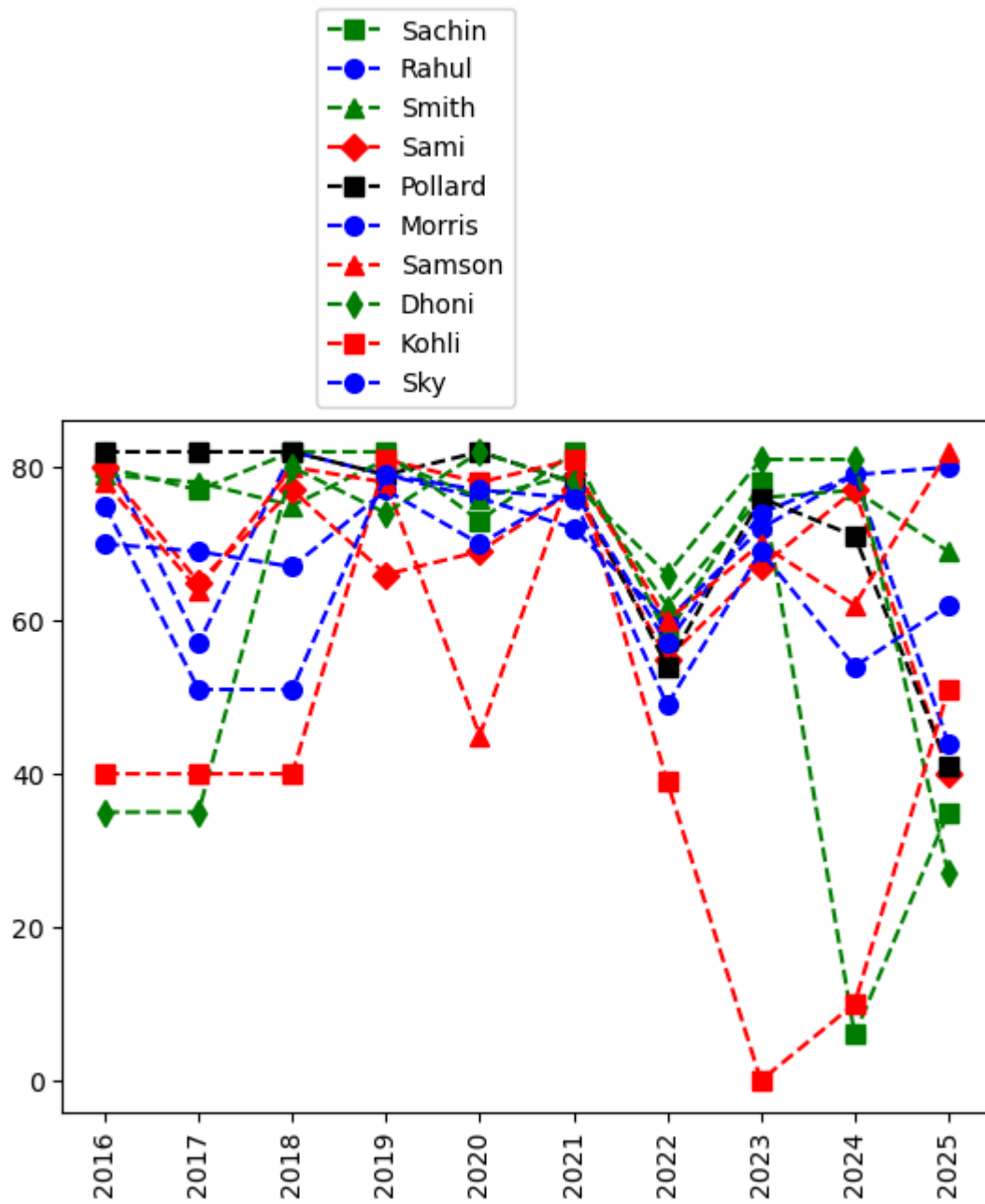
```
In [57]: plt.plot(Salary[0],color = 'r',marker= '.',ls = '--',ms=10,label = Players[0])
plt.plot(Salary[1],color = 'b',marker= 'o',ls = ':',ms=8,label = Players[1])
plt.plot(Salary[2],color = 'g',marker= '^',ls = '-.',ms=8,label = Players[2])
plt.plot(Salary[3],color = 'y',marker= 'd',ls = '-.',ms=9,label = Players[3])
plt.xticks(list(range(0,10)),Sdict,rotation='vertical')
plt.legend(loc = 'upper right',bbox_to_anchor=(1,0))
plt.show()
```



```
In [58]: plt.plot(Games[0], c='Green', ls = '--', marker = 's', ms = 7, label = Players[0])
plt.plot(Games[1], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[1])
plt.plot(Games[2], c='Green', ls = '--', marker = '^', ms = 7, label = Players[2])
plt.plot(Games[3], c='Red', ls = '--', marker = 'D', ms = 7, label = Players[3])
plt.plot(Games[4], c='Black', ls = '--', marker = 's', ms = 7, label = Players[4])
plt.plot(Games[5], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[5])
plt.plot(Games[6], c='red', ls = '--', marker = '^', ms = 7, label = Players[6])
plt.plot(Games[7], c='Green', ls = '--', marker = 'd', ms = 7, label = Players[7])
plt.plot(Games[8], c='Red', ls = '--', marker = 's', ms = 7, label = Players[8])
plt.plot(Games[9], c='Blue', ls = '--', marker = 'o', ms = 7, label = Players[9])

plt.legend(loc = 'lower right',bbox_to_anchor=(0.5,1) )
plt.xticks(list(range(0,10)), Sdict,rotation='vertical')

plt.show()
```



In []: